

A kinematic comparison of ergometer and on-water rowing

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ABSTRACT

The purpose of this investigation was to compare on-water rowing with ergometer rowing by identifying the important kinematic variables, using vector loop analysis. Two hypotheses were investigated that provided evidence that rowing ergometers are valid simulators of actual rowing and that the trunk segment is the major contributor to the total linear oar velocity. Thirty subjects were filmed rowing on-water and on an ergometer. The film was digitized for X and Y coordinates of the body joint locations for a cycle of rowing from catch to finish. A model of rowing was designed such that the rowing movement could be defined as comprising two closed vector loops. The digitized data were fitted to the vector loop model to derive kinematic variables from the drive phase of rowing. Five kinematic variables that described the contributions of five body segments to the total linear oar velocity were analyzed statistically to find differences in the patterns of on-water and ergometer rowing. The results indicated that the kinematics of the upper arm and forearm segments were significantly different in the two types of rowing. These differences were of minor importance because of the small contributions made by the arms at the catch. The differences at the finish were a result of the lifting of the oar from the water not exhibited in ergometer rowing. The vector loop analysis provided the major finding that the trunk segment contributed significantly more to the drive phase portion of the rowing cycle for both on-water and ergometer rowing although the legs initiated the drive phase. Despite the differences that occurred in the arm motion at the beginning and end of the drive phase, most of the kinematic variables were similar throughout the drive phase, particularly for the legs and trunk. The objective of developing a biomechanical test that uses kinematic measures obtained

on an ergometer is supported by the findings of this study, and such a study would supplement physiologic measures used in the selection of rowers.

Rowing is a sport that demands a high technical skill level as well as strength and other physiologic requirements. Training programs are often designed to include on-land conditioning with rowing simulators that approximate the rowing motion. The popularity of these training devices, which exist in several designs, is such that there are now "indoor rowing olympics" held on computerized rowing simulators.² These machines have an advantage over on-water rowing in that they are stationary, they do not involve the elements of drag from water, and they can measure work expenditure to convey an instant feedback to the athlete.

Although off-water rowing simulators, also known as rowing ergometers, have been used extensively for the physiologic testing of rowers,^{6,7} little evidence in the literature exists to support the belief that ergometer rowing replicates the same patterns of motion as used in on-water rowing.^{7,9,10} The United States Rowing Association uses as part of their selection process of candidates for the national rowing teams, physiologic measurements obtained from rowing ergometer tests. At present, there is no biomechanical test designed to evaluate rowing kinematics. A rowing ergometer capable of simulating the biomechanics of on-water rowing would be a valuable tool for use in team selection, training, and technical coaching of rowers.

The rowing stroke consists of a cyclic sequence of events that includes the catch, the drive phase, the finish, and the recovery. The catch occurs as the oar is placed quickly in the water and is followed by the drive phase, which includes the movement of the oar through the water.^{4,8,12} The finish occurs as the oar is withdrawn from the water and is followed by the recovery as the oar is moved through the air and is prepared to reenter the water. This study deals only with the analysis of the drive phase, because it is the drive phase that best allows an understanding of how the rower's shell

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is accelerated through the water. The end points of the rowing stroke are the catch and the finish, and these points define the start and end of the drive phase in one cycle of rowing.

The literature is inconclusive concerning an accepted sequence of body actions for initiating the drive phase in the rowing stroke. Edwards¹ described the trunk-thigh angle relationship as one of knee extension as well as trunk extension. Contraction of the hamstrings causes backward rotation of the trunk about the hips and increases the thigh-trunk angle. However, it is possible to increase the thigh-trunk angle by fixing the trunk and lowering the thighs by extension of the knees. If the quadriceps alone extend the knee, this extension will merely push the slide away to the backstop. If the knee is not extended, contraction of the hamstrings will cause the trunk to pivot about the hip joint. For proper opening of the trunk-thigh angle, the quadriceps and hamstrings should work simultaneously, with the quadriceps extending the knee and the hamstrings rotating the trunk. Improper use of the hamstrings results in "bum-shoving" or "shooting the tail" (the action of knee extension without the oar actually being pulled through the water).

Williams¹² determined that the speed of the blade relative to the shell during the drive phase is closely related to the speed of the shell through the water. Williams believed that the drive is initiated by the motion of the trunk and suggested that the proximal joint must be stabilized before distal joints are moved so that the sequence of the drive phase is an extension developing from the hips. The thigh-trunk angle must be the first to be opened, which explains why the initial drive must come from the trunk and not the legs.

Rosenberg (United States National Team Rowing Coach, 1975 to 1976) argued (unpublished data¹⁰) that if the stroke is performed properly, the legs accelerate the shell when it is at its lowest velocity and that then the lower trunk, midtrunk, arm, and wrist muscles become involved as the velocity of the shell increases. Martin and Bernfield⁶ argued that the legs initiate the drive and are the prime source of power. Nelson and Widule¹⁰ obtained results confirming that because of knee extension, the legs were the primary force-producing segments of the body in the early stage of the drive phase.

Vector loop modeling has seen limited application to the study of human movement.^{5,11} It has been frequently used for the study of mechanism analysis. Hall³ used its principles to develop the concept of kinematic coefficients where the motion of a mechanism can be expressed in terms of reference positions. The term "kinematic coefficient" represents the derivative of a motion variable with respect to an input or reference variable and allows for an understanding of how motion occurs within the framework of a body or machine.

There are two important purposes to this paper. The first purpose was to compare the on-water kinematics of the drive phase of the rowing cycle to the kinematics obtained on a dry land simulator, the rowing ergometer. There is little indication in the literature of this having been previously

attempted. It was hoped that by testing the hypothesis that ergometer rowing simulates on-water rowing, the results would support a proposed biomechanical test composed of kinematic variables to train and select future rowers. A second purpose of the study was to develop a vector loop model of rowing that would provide the framework for a comparison of on-water and ergometer rowing kinematics in order to determine the correct sequence of actions in the drive phase of sweep oar rowing.

MATERIALS AND METHODS

The subjects were 30 male candidates invited to the U.S. National Heavyweight Rowing Team selection camp of 1982. A Stanford Rowing Ergometer (Robert Austin Co.,) designed to simulate the technique and feel of on-water sweep rowing in an eight-oared shell, was used for the study. All of the subjects were filmed rowing on-water and on the rowing ergometer while they rowed at 30 to 31 strokes per minute and were instructed to use the same technique throughout the study.

All filming (ergometer and on-water) was done using a Milliken, Model DBM55 (D.B. Milliken Co., Arcadia, CA), 16 mm camera with a 50 mm lens and a 72° shutter with a film speed of 50 frames/sec that was verified by an internal pulse generator set at 10 pulses per second. For the ergometer filming, the camera was positioned at right angles to the plane of motion at a distance of 15 m. A board marked with a distance of 1 m was filmed in the location of the rower's body to determine the conversion factor from film length to actual linear length.

The on-water filming session involved mounting the camera on a motor-propelled launch that traveled at a comparable speed to the moving crew of eight rowers. The distance between the launch and the moving crew was approximately 20 m. The camera was focused on only one rower at a time, and the film was exposed for approximately six consecutive strokes of each rower.

From the processed film, every frame of the drive phase of one stroke was examined; sample sizes ranged from 40 to 62 frames. For each frame of film, nine points were plotted (toe, ankle, knee, hip, shoulder, elbow, wrist, oar handle, and head). The location of the seat was assumed to be represented by the position of the hip joint. The X and Y coordinates of these locations were assigned by the use of a Numonics digitizer (Numonics Corp., Lansdale, PA) linked with the Purdue Remote Control On-Line System (PROCSY) at the Biomechanics Laboratory, Lambert Gymnasium, Purdue University, and were subsequently stored as computer data files for each subject.

The data file coordinates were corrected for the inherent noise (error) by digital filtering to smooth the displacement data before the computation of kinematic information. Selection of filter coefficients corresponding to the fourth harmonic level (cut-off frequency) were used in a second-order Butterworth recursive digital filter to model the raw data.¹⁶ It was determined that this filter resulted in accept-

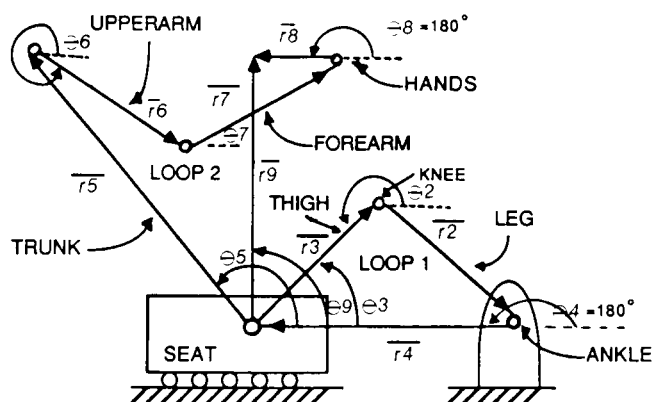


Figure 1. Schematic vector loop design of rower.

able values when five extra frames were included before and after the end points of the drive phase. These extra frames were then deleted after the filtering process before calculation of the displacement and velocity characteristics.

A FORTRAN program was written that calculated, for each frame of filtered film data, the specific angular measurements each body segment made with respect to the horizontal plane of reference. The angular displacements and velocities were calculated using these measurements. A total of 30 sets of data were obtained that matched the

subjects for on-water and ergometer rowing. There were a minimum of two and a maximum of four trials per subject of digitized ergometer and on-water rowing data, and the average score of each subject's trials was used for comparison between subjects. The data of filtered coordinates, angular displacements, and angular velocities were then fitted to a vector loop model of a rower (Fig. 1).

The vector loop model allows for a kinematic analysis to study the state of motion (position, velocity, and acceleration) of each body segment as represented by linkages within a vector loop. In this study, the problem was to decide which kinematic variables were of primary interest to compare ergometer rowing with on-water rowing. A random selection of kinematic variables would likely cause omission of some of the important variables or cause inclusion of too many variables to satisfy statistical criteria. The vector loop model eliminates this problem and, through a process of solving closed position and velocity vector equations, the kinematics of each body segment are directly related to the relative velocity of the hand, seat, and shell. To use the vector loop model of a rower (Fig. 1) in this study, it was assumed that the motion of the body segments in rowing is two-dimensional. This assumption was deemed valid by several rowing authorities and crew members at Purdue University. The tendency of the elbows to move away from the sagittal plane was assumed to be of minor consequence and did not present

TABLE 1
Comparison of mean contribution of body segment velocities ($\Sigma \text{VELVAR}(N) = 1.0$) and relative hand and seat velocity ($\Sigma \text{VEL}(N) = 1.0$) to the total linear oar velocity

	Percent of drive phase								
	10	20	30	40	50	60	70	80	90
Ergometer									
Trunk									
VELVAR1	0.249	0.282	0.293	0.356	0.474	0.604	0.626	0.530	0.387
Upper arm									
VELVAR2	0.002	0.006	0.046	0.043	0.052	0.107	0.178	0.192	0.118
Lower arm									
VELVAR3	0.071	-0.005	-0.001	0.001	0.003	0.016	0.076	0.265	0.527
Lower leg									
VELVAR4	0.678	0.551	0.479	0.445	0.366	0.227	0.107	0.017	-0.02
Upper leg									
VELVAR5	0.150	0.165	0.182	0.155	0.105	0.046	0.012	-0.004	-0.01
Seat									
VEL4	0.828	0.716	0.661	0.600	0.471	0.273	0.119	0.013	-0.03
Hand									
VEL8	0.171	0.284	0.339	0.399	0.529	0.727	0.881	0.987	1.03
Water									
Trunk									
VELVAR1	0.339	0.391	0.370	0.390	0.451	0.556	0.584	0.556	0.482
Upper arm									
VELVAR2	-0.208	-0.401	-0.166	0.011	0.072	0.072	0.187	0.287	0.435
Lower arm									
VELVAR3	0.005	0.015	0.013	0.016	0.016	0.012	0.036	0.051	0.026
Lower leg									
VELVAR4	0.762	0.708	0.511	0.397	0.375	0.295	0.166	0.098	0.056
Upper leg									
VELVAR5	0.102	0.193	0.267	0.176	0.107	0.060	0.027	0.008	0.002
Seat									
VEL4	0.959	0.981	0.772	0.581	0.462	0.355	0.193	0.105	0.056
Hand									
VEL8	0.038	0.012	0.229	0.417	0.538	0.645	0.807	0.895	0.944

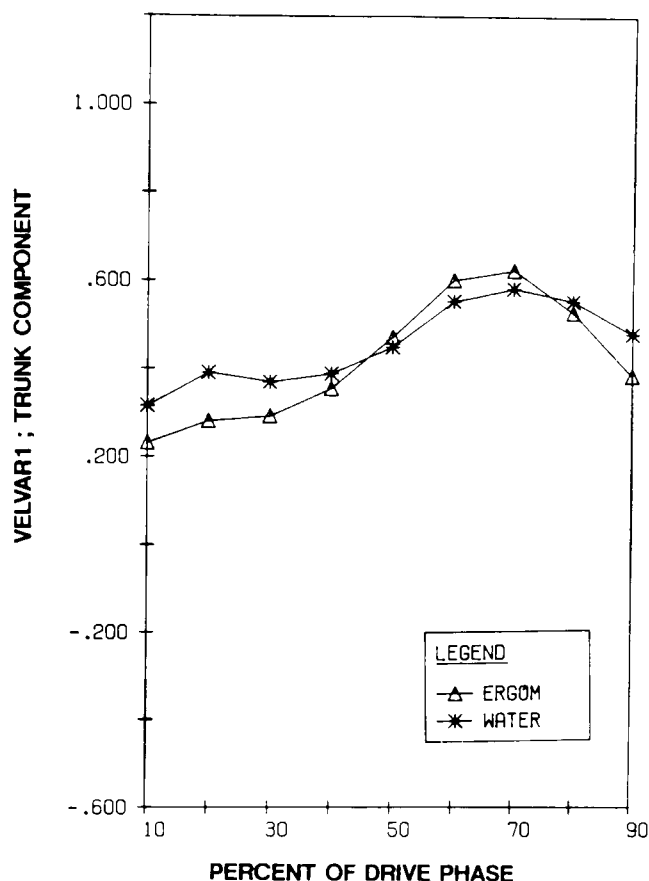


Figure 2. Comparison of trunk component velocity (VELVAR1) means for on-water rowing and ergometer rowing.

a serious problem to the study. The rower's two-dimensional side view was constructed into two closed vector loops consisting of the interconnected joint locations of the rower's body. The vectors were chosen and arranged so that the variables (lengths and angles) would provide the most useful information in the analysis. For the vector loop representing the lower body shown in Figure 1:

$$\bar{r}_3 + \bar{r}_2 + \bar{r}_4 = 0 \quad (1)$$

where $\bar{r}_2$ = length of leg segment

$\bar{r}_3$ = length of thigh segment

$\bar{r}_4$ = distance from ankle to seat

For the vector loop representing the upper body in Figure 1:

$$\bar{r}_5 + \bar{r}_6 + \bar{r}_7 + \bar{r}_8 - \bar{r}_9 = 0 \quad (2)$$

where $\bar{r}_5$ = length of the trunk segment

$\bar{r}_6$ = length of the upper arm segment

$\bar{r}_7$ = length of the lower arm segment

$\bar{r}_8$ = distance from hands to perpendicular through the seat

$\bar{r}_9$ = distance from seat to intersection with $\bar{r}_8$

For the lower body vector loop in Figure 1, kinematic coefficients were calculated for: 1) the derivative of the upper

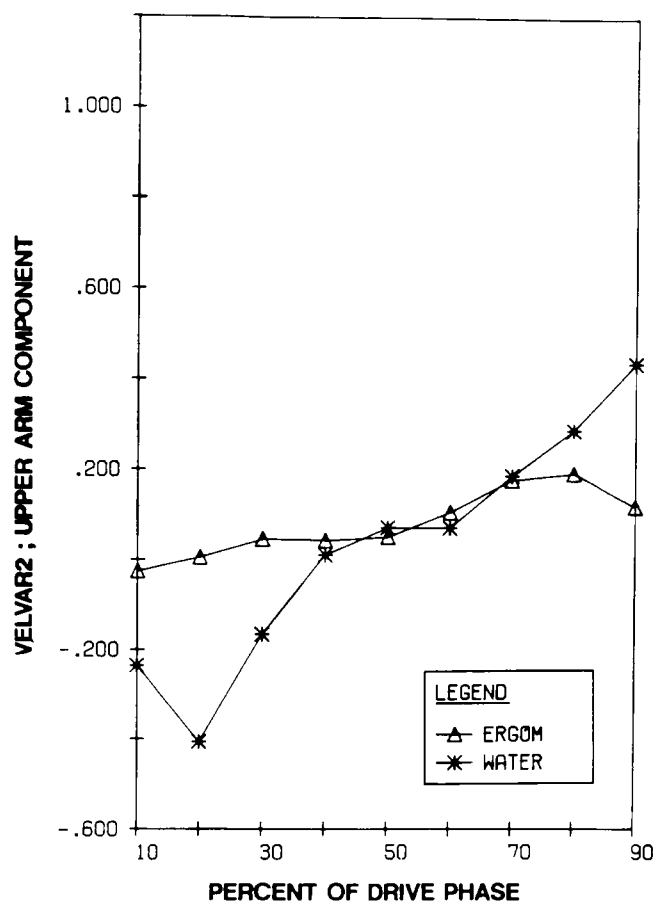


Figure 3. Comparison of upper arm component velocity (VELVAR2) means for on-water rowing and ergometer rowing.

leg angle (θ_3) with respect to the derivative of the lower leg angle (θ_2), and 2) the derivative of the seat to ankle distance ($\bar{r}_4$) with respect to the derivative of the lower leg angle (θ_2).

For the upper body vector loop in Figure 1, kinematic coefficients were calculated for: 1) the derivative of the upper arm angle (θ_6) with respect to the derivative of the trunk angle (θ_5), 2) the derivative of the lower arm angle (θ_7) with respect to the derivative of the trunk angle (θ_5), and 3) the derivative of the hands-to-seat horizontal distance ($\bar{r}_8$) with respect to the derivative of the trunk angle (θ_5).

The net result of determining the kinematic coefficients for the two vector loops was that the total linear oar velocity could be expressed in terms of five velocity variables:

- 1) VELVAR1 (trunk component velocity)
- 2) VELVAR2 (upper arm component velocity)
- 3) VELVAR3 (lower arm component velocity)
- 4) VELVAR4 (lower leg component velocity)
- 5) VELVAR5 (upper leg component velocity)

These five variables were the major kinematic variables of interest that were investigated in this study. Two additional kinematic variables were also investigated, VEL4 and VEL8, which represented the normalized relative seat veloc-

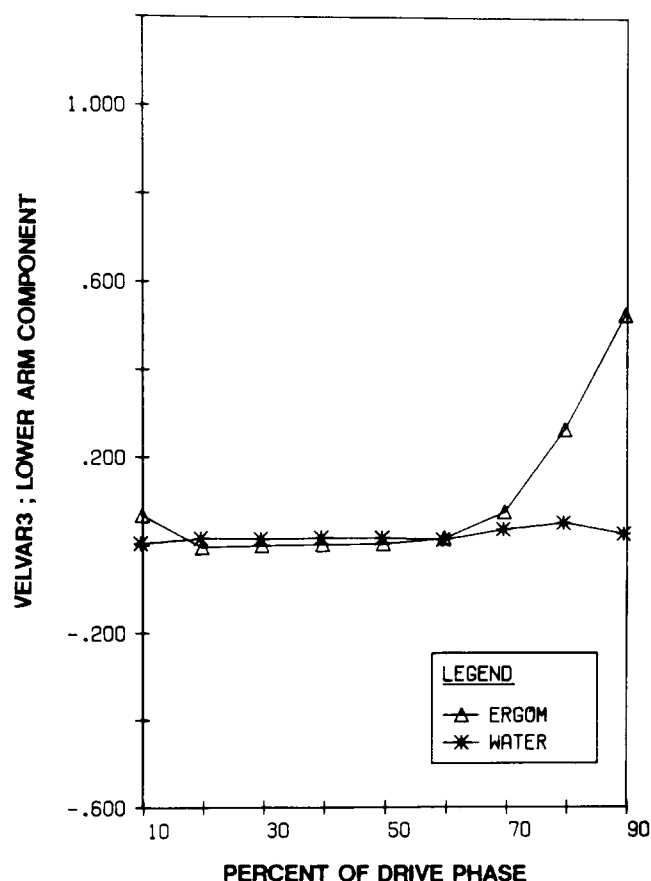


Figure 4. Comparison of lower arm component velocity (VELVAR3) means for on-water rowing and ergometer rowing.

ity and normalized hand velocity, respectively. Normalization of all of the kinematic variables was done by dividing the kinematic coefficient containing variables (VELVAR1 to VELVAR5 and VEL4 and VEL8) by the total linear oar velocity. This produced a linear regression model where the contribution of each kinematic variable (VELVAR1 to VELVAR5) could be correlated to the total linear oar velocity:

$$\text{VELVAR1} + \text{VELVAR2} + \text{VELVAR3} + \text{VELVAR4} + \text{VELVAR5} = 1$$

The kinematic variables VEL4 and VEL8 represented the correlation of the total lower body velocity and the total upper body velocity to the total linear oar velocity:

$$\text{VEL4} + \text{VEL8} = 1$$

To compare the obtained kinematic variables for the two types of rowing, the statistical design used was a repeated measures, two factors, analysis of variance (ANOVA) with the subjects nested within the factors. The two factors were the type of rowing and the percent of drive phase.

The factor of percent of drive phase compared each dependent kinematic variable at each 10% of the drive phase for nine successive intervals, excluding the end points of 0% and 100%. The percent of drive phase was determined by

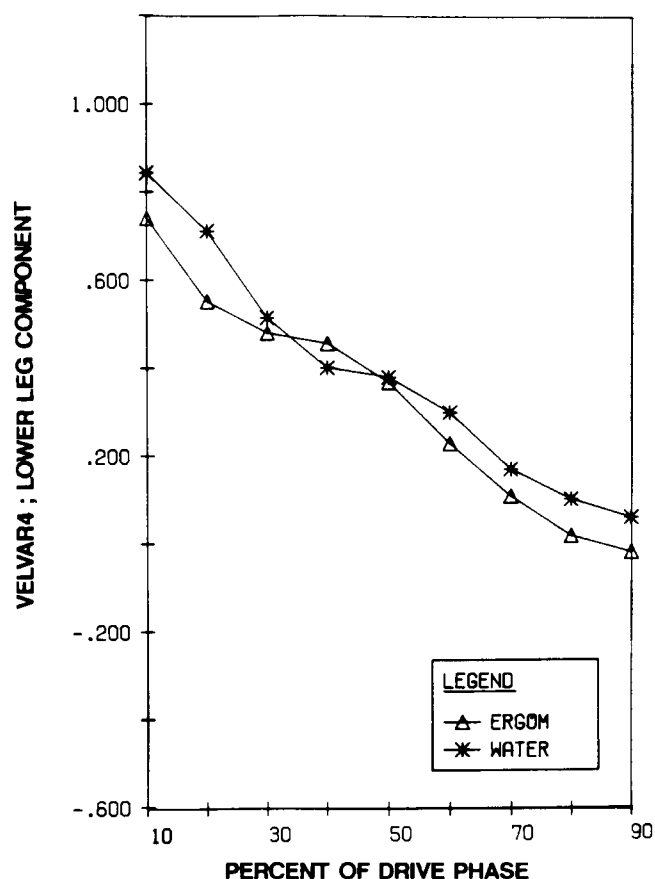


Figure 5. Comparison of lower leg component velocity (VELVAR4) means for on-water rowing and ergometer rowing.

dividing the frame number by the total number of frames and multiplying by 100. Each 10% level value was obtained by selecting the measure closest to 10% or a multiple of 10%. Therefore, kinematic measures were obtained at 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, and 90% of the drive phase.

The ANOVA tested for the presence of interaction between the type of rowing and the percent of the drive phase. A post hoc Scheffe test was used for determining which pairs of group means were statistically different when the kinematic variables had a positive interaction.

RESULTS

Table 1 presents a list of the group means for all of the kinematic variables at each corresponding 10% level of the drive phase for both types of rowing. It represents a comparison of 30 subjects between on-water and ergometer rowing. These scores are graphically represented in Figures 2 to 8.

It was apparent that the relative seat velocity and the relative hand velocity were not constant throughout the drive phase of rowing (Fig. 7 and 8). The contribution made by the relative seat velocity to the total linear oar velocity

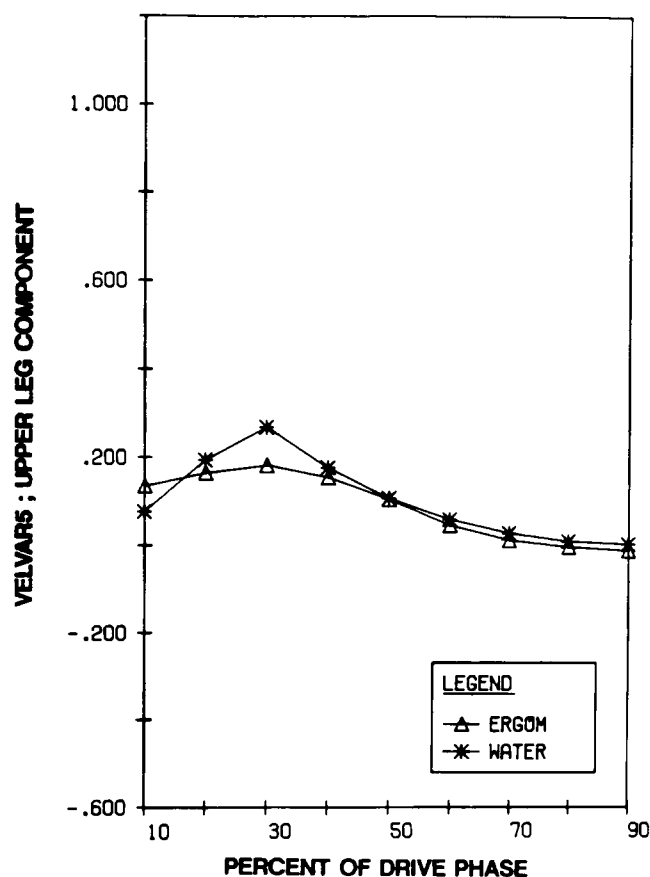


Figure 6. Comparison of upper leg component velocity (VELVAR5) means for on-water rowing and ergometer rowing.

was much greater than the contribution made by the relative hand velocity during the first 40% of the drive phase for both types of rowing. The remaining 60% of the drive phase showed a greater contribution from the relative hand velocity to the total linear oar velocity for both types of rowing.

The contribution from the lower leg velocity to the total linear oar velocity was greater than the contribution of the upper leg velocity at all percent levels of the drive phase for both types of rowing (Figs. 4 and 5).

The trunk velocity contribution to the total linear oar velocity maximized at approximately 70% of the drive phase for both ergometer and on-water rowing (Fig. 2).

There are few observable differences between the two types of rowing within Figures 2 to 8, which indicates support for the hypothesis that ergometer and on-water kinematics are similar. The results of the ANOVA indicated a significant ($P < 0.05$) interaction for the type of rowing versus percent of drive phase for kinematic variables VELVAR2 and VELVAR3. These variables were derived from the upper body vector loop and represent the upper arm contribution and lower arm contribution to the total linear oar velocity, respectively. A post hoc Scheffe test on these variables indicated that the group means for ergometer versus on-water rowing were significantly different at 10%,

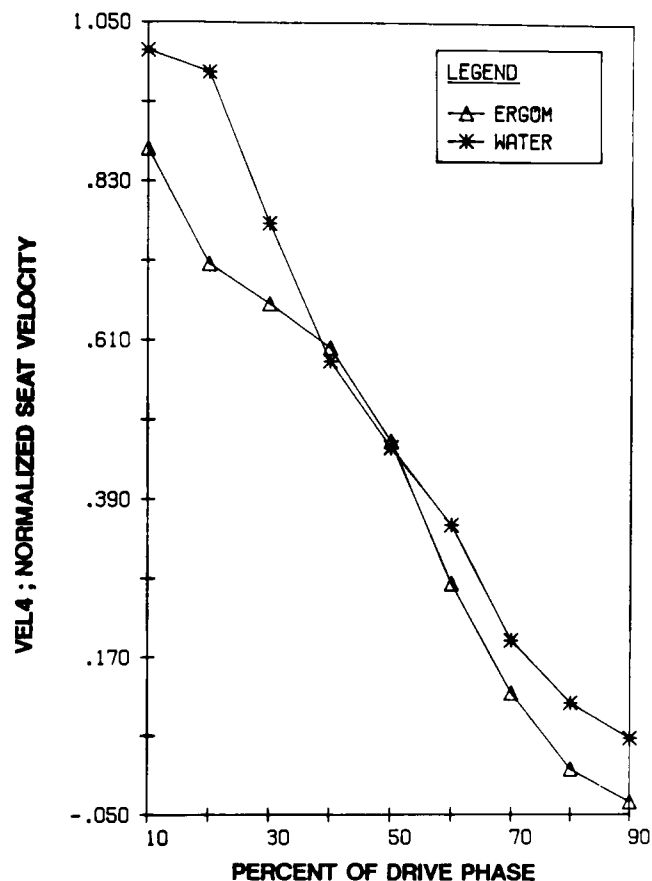


Figure 7. Comparison of normalized relative seat velocity (VEL4) means for on-water rowing and ergometer rowing.

20%, 30%, and 90% of the drive phase for VELVAR2 (Fig. 3), and at 80% and 90% of the drive phase for VELVAR3 (Fig. 4).

DISCUSSION

The vector loop model of rowing helped to identify the contributions made by the body segments to the total linear oar velocity. The subjects of this study exhibited similar kinematics between on-water and ergometer rowing for both the trunk and leg components. They also demonstrated that these components are the prime contributors to the drive phase of sweep oar rowing. It is apparent from comparing Figure 5 with Figures 2 and 6, that the lower leg is responsible for initiating the drive phase in sweep oar rowing. Somewhere between 40% and 50% of the drive phase, the trunk contribution equals and then surpasses the contribution made by the lower legs.

The statistical differences between the two types of rowing for the upper and lower arm components occurred only at the beginning and end of the drive phase (Figs. 3 and 4). The differences at the beginning of the drive phase can be explained by realizing that slippage of the oar in the water creates a situation in which the arms compensate for the nonstationary fulcrum of the oar in the water. The catch

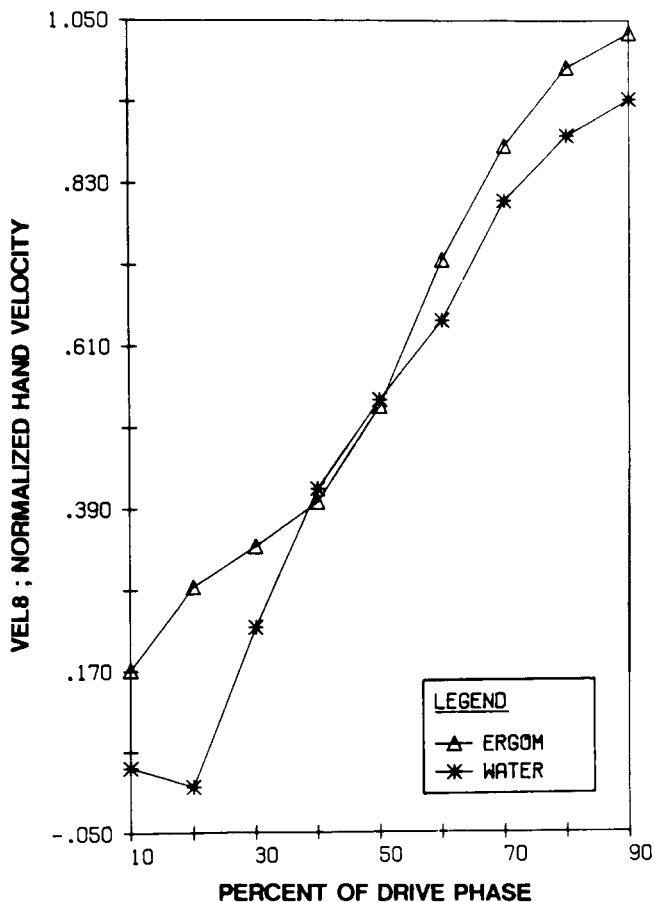


Figure 8. Comparison of normalized relative hand velocity (VEL8) means for on-water rowing and ergometer rowing.

position in ergometer rowing occurs around a fulcrum that is fixed and remains stationary. Therefore the leverage of pull on the oar using the arm segments would tend to be greater for ergometer rowing than for on-water rowing. If technological changes in design were to be made to the ergometer, it might be feasible to suggest incorporating a spring mechanism in the fulcrum to better simulate the slippage of the oar in on-water rowing.

The arm segment motions (VELVAR2 and VELVAR3) were also significantly different at the end of the drive phase ($P < 0.05$). This may have been the result of a reduced tendency to "feather" or lift the oar at the finish in ergometer rowing, which is essential in on-water rowing. For practical purposes, ergometer rowing should emphasize the motions of feathering the oar or hyperextending the wrists at the finish. In this study, the results indicated that the subjects did not decelerate the lower arm movements at the end of the drive phase for ergometer rowing and generated a greater relative hand velocity at the finish.

The fact that significant differences did occur between the two types of rowing for the arm segments should not be interpreted as grounds for rejecting the original hypothesis because the absolute measures of the arm component veloc-

ities were relatively small in comparison to the contributions made by the trunk and legs at the beginning of the drive phase. The significant differences obtained for the arms would appear to be of minor importance when consideration is given to the contributions made by the different body segments to the total linear velocity.

In conclusion, the vector loop analysis was able to generate important kinematic information for the motions of rowing and proved that the trunk segment of the body is the major contributor to generating total linear oar velocity. This is in contrast to the research of Martin and Bernfield,⁶ who stated that the legs are the prime source of power.

The hypothesis that ergometer rowing simulates on-water rowing was supported from the findings of this study. Despite some differences that occurred in arm motion at the beginning and end of the drive phase, the kinematic variables of the legs and trunk were similar between on-water and ergometer rowing. The objective of developing a biomechanical test that uses kinematic measures obtained on an ergometer would appear to be valid in supplementing physiological measures used in the selection of rowers.

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